Richard G. Cosentino

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Professional Employment:

Dates	Position	Location
2021-present	Senior Staff Scientist	Space Telescope Science Institute Baltimore, MD
2019-2021	Postdoctoral Research Associate	Department of Astronomy Univ. of Maryland - CRESST-II
2017-2019	NASA Postdoc Program	NASA Goddard Space Flight Center (GFSC) Greenbelt, MD
2016-2017	Grote Reber Doctoral Fellow	National Radio Astronomy Observatory (NRAO) Socorro, New Mexico
2010-2017	Research and Teaching Assistant	Physics Department, New Mexico Tech (NMT) Socorro, NM
2005-2010	Electrical Engineer	Integrated Building Concepts, IBC LLC Buffalo, NY

Educational Background:

Date	Degree	University
2017	Ph.D. Physics - Astrophysics Dissertation Title: "Investigating Jovian Tropospheric and Stratospheric Vertical Structure with Numerical Models and Radio Wavelength Observations"	New Mexico Institute of Mining and Technology (NMT)
2005	B.S. Engineering Physics and Electrical Engineering	State University of New York, Buffalo

Areas of Research:

- Optical and near-infrared imaging, spectroscopy, and instrumentation; experience specifically with solar system small bodies / asteroids.
- Atmospheric Dynamics - jet streams, convection, waves, and turbulence to interpret vertical wind structure, and large-scale 3D circulation with comparative planetology and climatology; Jupiter, Saturn, Earth, Titan, Uranus, Neptune, and exoplanets.
- Multi-wavelength observations of planetary atmospheres with HST, IRTF, VLA, and ALMA. Expertise with submm and radio wavelength interferometry.

Mission Experience:

- Nancy Grace Roman space telescope mission instrument scientist on the Wide Field Instrument (WFI), which is an array of (18) 4k near infrared detectors with a large FOV. Developing and implementing the calibration reference file pipeline system. Coordinate with various working groups and sub-systems in reference file creation according to the calibration plan with ground and in-flight systems, scheduling and operations, and data management. Other duties include documentation, community coordination of survey definition and project teams, and in-flight operations of observations through Multi-Accumulation Tables (i.e. integration instructions).
- OSIRIS-REx mission instrument scientist on the visible and infrared spectrometer (OVIRS). Observation mission planning, technical support, and data analysis. Trending analysis of instrument internal calibration sources, health, performance, and functionality.

First Authored Publications:

1. **Cosentino, R.G.**, Greathouse, T., Simon, A.A, Morales-Juberías, R. et al. "The Effects of Waves on the Meridional Thermal Structure of Jupiter's Stratosphere", Planetary Science Journal, Nov 2020.

2. **Cosentino, R.G.**, Simon, A.A, Morales-Juberías, R., "Jupiter's Turbulent Power Spectra from Hubble Space Telescope", Journal of Geophysical Research - Planets, Apr 2019.

3. **Cosentino, R.G.**, Morales-Juberías, R., Greathouse, T., Orton, G., Johnson, P., Fletcher, L.N., Simon, A.A, "New Observations and Modeling of Jupiter's Quasi-Quadrennial Oscillation", Journal of Geophysical Research - Planets, Dec 2017.

4. **Cosentino, R.G.**, Butler, B., Sault, R., Morales-Juberías, R., Simon, A., de Pater, I., "Atmospheric waves and dynamics beneath Jupiter's clouds from radio wavelength observations", Icarus, Vol 292, 168-181, Aug 2017.

5. **Cosentino, R.G.**, Simon, A., Morales-Juberías, R., Sayanagi, K.M., "Observations and Numerical Modeling of the Jovian Ribbon", Astrophysical Journal Letters, 810, Aug 2015.

Successful Observing Proposals as PI:

Telescope	Title	Time	Status
ALMA Cycle 13	ID Monitoring Giant Planet High-Altitude Jet Streams	2.6 hrs	Submitted New Project Observations
ALMA Cycle 11	ID 2024.1.01691.S: Monitoring Giant Planet High-Altitude Jet Streams	2.6 hrs	Continuing Analysis
ALMA Cycle 11	ID 2024.1.00104.S: Uranus' Polar Bullseye	3.2 hrs	Analyzing Data
ALMA Cycle 10	ID 2023.1.01703.S: Monitoring Giant Planet High-Altitude Jet Streams	2.6 hrs	Data Analyzed
ALMA Cycle 9,10	ID 2022.1.00548.S: Neptune's High-Altitude Wind Chill from ALMA	1.8 hrs	Preparing Publication
ALMA Cycle 9	ID 2022.1.01625.S: Monitoring Giant Planet High-Altitude Jet Streams	2.6 hrs	Data Analyzed
IRTF 2022	ID 2022B096: Titan's Stratospheric Winds from an Occultation before Equinox	Half night	Published
HST 2016	ID 14585: Understanding Jupiter's Cloud Scale and Energy Spectra from Archival Data	Archival	Published
VLA 2015	VLA 15A-320: Studying Jupiter's Deep Atmosphere, Waves and Turbulence	10 hrs	Published

Successful Funding Proposals as PI:

Source	Title	Duration	Status
STScI Internal	Neptune Wind Chill (0.2 FTE)	1 yr	Completed, Report Submitted
STScI Internal	Asteroids are not all Cosmic Rays for Roman (0.1 FTE)	1 yr	Preparing Publication

First Authored Technical Reports:

1. **Cosentino, R.G.**, McDonald, M., 'RTB Deliveries to CRDS Through Commissioning and into Operations", *in preparation.*

2. **Cosentino, R.G.**, Schultz, W., Sappington, B. Gomez, S.'Coordination of RFP Quality Control with RDMT Reef", *in review.*

3. **Cosentino, R.G.**, Desjardin, T., 'The Wide Field Instrument Reference File Pipeline", Technical Report Roman-STScI-000811 Baltimore: STScI, 2026.

4. **Cosentino, R.G.**, Sappington, B., Desjardin, T., 'The Wide Field Instrument Calibration Reference File Pipeline Performance Testing of the In-Flight Dark Calibration Plan", Technical Report Roman-STScI-000678, Baltimore: STScI, 2024.

Publications in preparation:

1. **Cosentino, R.G.**, Loughran, S., Simon, A.A., et al. "Neptune's stratospheric winds and storms from ALMA.". (ApJ Letters)

2. **Cosentino, R.G.**, Simon, A.A., et al. "Spatial and Temporal Variability in Gas Giant Upper Atmospheric Winds". (Planetary Science Journal)

3. **Cosentino, R.G.** "A Review of Stratospheric Variability in the Solar System". (Journal of Geophysics)

Appointments:

1. NRAO Science Review Panel on Solar System, Stars, and Planetary Systems, 2-year term as member. July 2024-2026

Selected Co-Authored Publications:

1. Holler, B., **Cosentino, R.G.**, et al. "The Roman Space Telescope as a Planetary Defense Asset". Publications of the Astronomical Society of the Pacific, 2025.

2. Cordiner, M., Thelen, A., Cavali, T., **Cosentino, R.G.**, et al. "Atacama Large Aperture Submillimeter Telescope (AtLAST) Science: Planetary and Cometary Atmospheres". Open Research Europe, 2024.

3. Dahl, E., Rowe-Gurney, N., Orton, G., Brueshaber, S., **Cosentino, R.G.**, Cordiner, M., et al. "Atmospheric Science Questions for a Uranian Probe". Space Science Reviews, 2024. 210:10.

4. Thelen, A., Nixon, C., **Cosentino, R.G.**, Cordiner, M., et al. "Variability of Titan's Mesospheric HCN and Temperature Structure from ALMA". PSJ, June 2022.

5. Morales-Juberias, R., Simon, A., **Cosentino, R.G.**, "Analysis of the long-term drift rates and oscillations of Jupiter's largest vortices". Icarus, Jan 2022.

6. Cordiner, M., Garcia-Berrioes, E., **Cosentino, R.G.**, Teanby, N.A., et al. "Rapid Variability of Titan's Thermospheric Jet Discovered using ALMA". ApJ Letters, Nov 2020.

7. Antunano, A., **Cosentino, R.G.**, Fletcher, L., Simon, A.A, et al. "Fluctuations in Jupiter's Equatorial Stratospheric Oscillation". Nature Astronomy, Jun 2020.

8. Giles, R., Greathouse, T., **Cosentino, R.**, Orton, G., Lacy, J. "Vertically-resolved observations of Jupiter's quasi-quadrennial oscillation from 2012-2019". Icarus, Jun 2020.

9. Simon, A., K. Donaldson Hanna, C. Drouet dAubigny, G. Poggiali, J. Emery, J. Brucato, **R. Cosentino**, et al. "OSIRISREx visible and nearinfrared observations of the Moon.", Geophysical Research Letters, Jun 2019.

10. Simon, A.A, Reuter, D., Gorius, N., Lunsford, A., **Cosentino, R.G.**, Wind, G., Lauretta, D., et al. "In-Flight Calibration and Performance of the OSIRIS-REx Visible and IR Spectrometer (OVIRS)", Remote Sensing, Sep 2018.

11. Simon, A.A, Tabataba-Vakili, F., **Cosentino, R.G.**, Beebe, R., Wong, M.H., Orton, G. "Historical and Contemporary Trends in the Size, Drift and Color of Jupiter's Great Red Spot", Astronomical Journal, Mar 2018.

12. Fletcher, L., Guerlet, S., Orton, G., **Cosentino, R.G.**, Fouchet, T., Irwin, P., Li, L., Flasar, M., Gorius, N., Morales-Juberías, R., "Disruption of Saturns quasi-periodic equatorial oscillation by the great northern storm", Nature Astronomy, Oct 2017.

13. Morales-Juberías, R., Sayanagi, K., Simon, A., Fletcher, L., **Cosentino, R.G.**, "Meandering Shallow Atmospheric Jet as a Model of Saturn's North Polar Hexagon", Astrophysical Journal Letters, Jun 2015.

Selected Co-Authored Technical Reports:

1. Desjardins, T., Betti, S., **Cosentino, R.G.**, et al. "Roman SOC Science Validation Report for SOC Release 2 DMS", Technical Report Roman-STScI-000619, Baltimore: STScI, 2024.

2. Petric, A., Casertano, S., Beaton, R., Betti, S., Bellini, A., **Cosentino, R.G.**, et al. "Approximating Uncertainties for Calibration Reference Files", Technical Report Roman-STScI-000571, Baltimore: STScI, 2024.

3. Desjardins, T., Bellini, A,. **Cosentino, R.G.**, Hoffmann, S., Snchez, J. "Roman SOC Science Validation Report for SOC Release 1 DMS", Technical Report Roman-STScI-000456, Baltimore: STScI, 2023.

4. Petric, A., Casertano, S., Wu, J., Beaton, R., Bellini, a., **Cosentino, R.G.**, et al. "Characterization of Roman Detectors Darks Data Acquired with Leach and ACADIA Controllers", Technical Report Roman-STScI-000396, Baltimore: STScI, 2023.

Conferences:

Meeting	Title
AGU 2025	(Poster) Neptunes Upper Atmospheric Winds and Vertical Wind Shear from ALMA Observations
DPS 2022	(Poster) The Roman Space Telescope Science Operations Center: Tools and Applications for Astrophysics and Planetary Science
AAS 2022	(Poster) The Roman Space Telescope Science Operations Center: Overview of Software and Data Simulation Tools
DPS 2020	(Poster) The Turbulence of Saturn's Northern Hemisphere from the Hubble OPAL campaign
AGU 2018	(Talk) Jupiter's Turbulent Power Spectra from Hubble Space Telescope
DPS 2018	(Poster) Comparing Thermal Structure of Anomalies in Jupiter's Stratosphere with IRTF and Gemini
DPS 2017	(Poster) Long Variations and Statistical Properties of Jupiter's Stratospheric Oscillation
DPS 2016	(Talk) Modeling Jupiter's Quasi Quadrennial Oscillation (QQO) with Wave Drag Parameterizations
DPS 2015	(Poster) Searching for atmospheric waves below the Jupiter's clouds using radio wavelength observation Session Chair - (400) Jovian Planet Atmospheres: Storms and Upper Atmosphere
DPS 2014	(Poster) Numerical modelling of planetary-scale waves on Jupiter
AGU 2013	(Poster) Atmospheric Oscillations in Gas Giant Planets
AGU 2012	(Poster) Simulations of the Jovian Stratosphere with diabatic heating and mechanical forcing terms

Teaching - NMT Physics:

Date	Course
F15	Teaching Assistant for PHYS 121: Physics I Recitation (mechanics)
F15, F12	Teaching Assistant for PHYS 121L: Physics I Laboratory (mechanics)
S14, F13, S13	Teaching Assistant for PHYS 122L: Physics II Laboratory (optics, EM) Updated and improved existing lab manual. Designed and added new labs.

Fall Semester - F Spring Semester - S

Workshops:

Date	Meeting Name	Role
Nov 2025	STScI/JHU Exoplanet Jamboree	Presenter, SOC, LOC
May 2024	Uranus Flagship: Investigating New Paradigms for Outer Planet Exploration Workshop	Presenter, Attendee
Sept 2020	TRAPPIST Habitable Atmosphere Intercomparison (THAI) Workshop	Attendee
June 2019	Planetary CubeSats/SmallSats Symposium	Attendee
May 2016	NRAO 15th Interferometry Synthesis Imaging Workshop	Volunteer
April 2015	NMT Graduate Student Association (GSA) Interdisciplinary Symposium on Astrobiology	Presenter, Attendee, Emcee and LOC
May 2014	NRAO 14th Interferometry Synthesis Imaging Workshop	Attendee
July 2013	NRAO 7th Single Dish Summer School	Attendee
June 2012	Stratospheric Processes And their Role in Climate (SPARC) in Climate (SPARC) Data Workshop	Attendee and LOC
May 2012	Making New Worlds: Atmospheric, Thermal and Astrobiological Interpretation of Exoplanets	Attendee

Outreach and Education:

Date	Activity	Location
2024	Howard County Astronomy Club Meeting	Columbia MD
2020	Frostburg State University Star Party	Frostburg MD
2019	Apollo Anniversary Celebration Volunteer	D.C.
2013-2016	Guide and Volunteer	VLA Site
2011-2016	Star Party Volunteer and Organizer at Etscorn Observatory	Socorro NM
2015	Year of the Dwarf Planet Lectures for Upward Bound Science Summer Program	Socorro NM
2011-2015	Science Fair Judge for State High, Middle and Elementary Schools	Socorro NM
2014	Division of Planetary Science Third Cohort of Astronomy Ambassadors	Tucson AZ

Selected Talks:

Date	Event	Title
2025	STScI/JHU Exoplanet Jamboree	"Time-Variable Stratospheres in the Solar System and Implications for Exoplanet Atmosphere Studies"
2024	JHU Howard County Astronomical Society	"Storms of Solar System Planets; Polar Cyclones to the Great Red Spot"
2021	University of Maryland PALS Seminar	"Convective storms, waves, and super-rotation in the atmospheres of Jupiter and Saturn"
2021	Goddard Space Flight Center Seminar	"How convection and storms influence Jupiter's stratospheric oscillation"
2019	Goddard Space Flight Center Seminar	"How Convection and Climate Impact Atmospheric Jets on Earth, Jupiter and Saturn"
2018	Earth Networks Lunch n' Learn	"Weather and the Atmospheres of Jupiter and Saturn"
2017	Goddard Space Flight Center Seminar	"Numerical Models and Convection for Jupiter's Clouds, Winds, and Temperatures"
2016	New Mexico Symposium (NRAO)	"Atmospheric waves and dynamics beneath Jupiter's clouds from radio wavelength observations"
2016	University of New Mexico Physics Colloquium	"The upgraded VLA looks for waves and dynamics beneath Jupiter's clouds"
2015	NMT Physics Colloquium	"Observations and Numerical Modeling of" the Jovian Ribbon"
2015	NMT GSA Astrobiology Symposium	"Evolution of the Inner Planet Atmospheres"
2013	NMT Geology Department Volcano Lunch Talk	"Volcanoes in the Solar System"

Honors/Awards:

Date	Description	Institution
2017	Petschek Award for Theoretical Physics	NMT
2016	Grote Reber Doctoral Fellowship	NRAO
2012	Leslie Fallon Teaching Award for Excellence in Physics Lab Education	NMT